

# **SEM–DSM Hybrid Method Manual**

A practical guide for localized 3-D teleseismic waveform modelling

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# Scope

This manual describes the practical workflow for running the SEM–DSM hybrid method implemented in this repository. The method combines the Direct Solution Method (DSM), which computes wave propagation in a 1-D spherical Earth model, with a localized 3-D SPECFEM3D simulation. The 1-D incident wavefield is injected into a small SEM box that contains the target 3-D heterogeneity, and the scattered wavefield recorded on an internal coupling boundary is extrapolated to teleseismic receivers with DSM Green’s functions.

The manual is written for users who want to run or adapt the example cases. The main user-facing workflow is controlled by a case directory. Its `DATA/` directory contains the configuration and Fortran model sources, including `model_tomography.f90` and `get_model.F90`; `auxiliary/` contains the helper programs; and the `step*.sh` scripts drive the workflow. Each case has a Step 0 script that activates its SPECFEM3D database-model sources before the remaining workflow is run.

# Chapter 1

## Introduction

### 1.1 Why a hybrid method

Fully global 3-D simulations at frequencies near 1 Hz are usually too expensive for routine teleseismic modelling. In many applications the structure of interest is localized: an ultra-low velocity zone near the CMB, topography on the ICB, or small-scale inner-core heterogeneity along one ray bundle. The SEM–DSM method exploits this localization. The global background propagation is handled by DSM in a 1-D Earth model, while the expensive 3-D calculation is restricted to a small box around the target region.

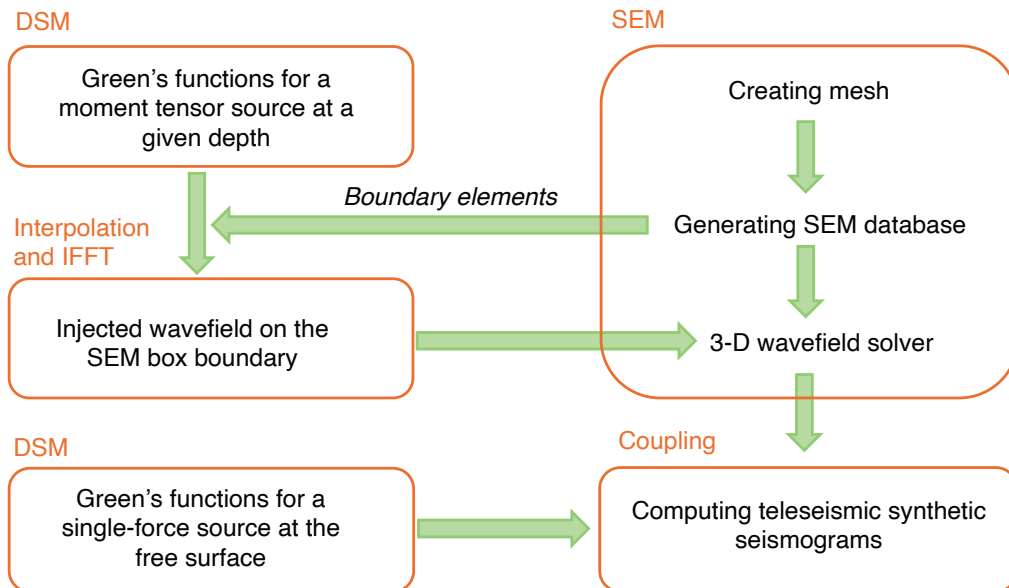


Figure 1.1: High-level workflow of the SEM–DSM hybrid method. DSM is used for source-to-box, box-to-receiver, and reference 1-D synthetics. The source-to-box DSM wavefield is converted to the injected wavefield used by SPECFEM3D. The SPECFEM3D boundary displacement and traction are combined with box-to-receiver DSM Green's functions to produce scattered teleseismic synthetics.

## 1.2 Coupling concept

Let the Earth be divided into an exterior domain  $V_1$ , which contains the earthquake source and teleseismic receivers, and a localized 3-D SEM domain  $V_2$ , which contains the target structure. The interface used for reconstructing the scattered wavefield is  $S_{12}$ . In the exterior domain the wavefield is represented by 1-D DSM Green's functions:

$$u_n^{\text{scat}}(\mathbf{x}, t) = \int_{-\infty}^{\infty} \int_{S_{12}} \left[ G_{ni}^{1D}(\mathbf{x}, t - \tau; \boldsymbol{\xi}, 0) T_i^{3D}(\boldsymbol{\xi}, \tau) - u_j^{3D}(\boldsymbol{\xi}, \tau) C_{jklm}^{1D}(\boldsymbol{\xi}) \frac{\partial G_{nm}^{1D}}{\partial \xi_l} n_k \right] dS d\tau. \quad (1.1)$$

The full teleseismic waveform is the sum of the DSM 1-D reference waveform and the scattered waveform from Eq. 1.1.

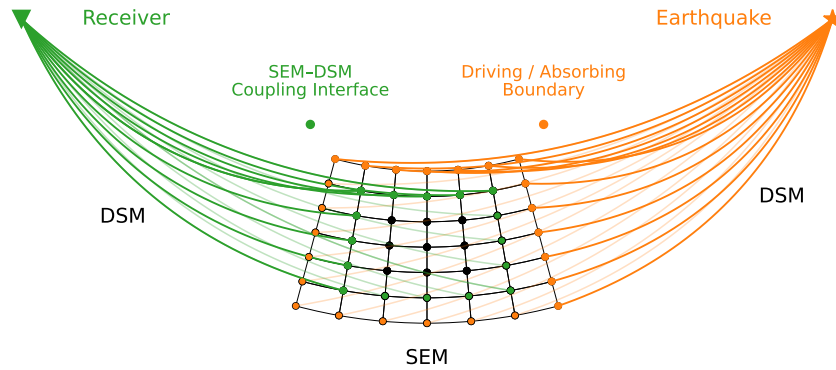


Figure 1.2: Schematic SEM-DSM coupling geometry. The outer SEM boundary is driven by the injected DSM incident wavefield. The internal coupling boundary provides the displacement and traction used in the representation theorem. Absorbing boundaries remove outgoing scattered waves.

## 1.3 Directory layout

A typical example directory has the following structure:

```
Example_case/
  DATA/
    Par_file_SEM_DSM
    CMTSOLUTION
    dsm_model_base
    model_tomography.f90
    get_model.F90
    tele_station.txt
  auxiliary/
    select_generate_databases_model.sh
    prepare_src_to_box.py
    prepare_box_to_recv.py
    prepare_injected_waves.py
    assemble_3d_sem_dsm_synthetic.py
  step0_prepare_*.sh
```

```
step1_specfem_mesh_database.sh
step2_3Dmodel_implementation.sh
step3_submit_specfem_mesh_gen.sh
step4_dsm_src_to_box.sh
step5_dsm_box_to_recv.sh
step6_1d_dsm_synthetics.sh
step7_extract_injected_waves.sh
step8_specfem_solver.sh
step9_coupling.sh
step10_assemble_3d_sem_dsm_synthetics.sh
```

Each case contains one `step0_prepare_*.sh` script. The descriptive suffix varies with the example because Step 0 may also prepare topography or volumetric heterogeneity files.

Steps 1–10 write most generated products under `WORK/`. Step 0 may also write case-specific structural inputs under `DATA/`. Large completed products, such as reusable DSM databases or final synthetics, are often saved as `OUTPUT_FILES_*` directories by individual examples.

## Chapter 2

# Compile Codes

### 2.1 Prerequisites

The code is intended for an MPI-enabled HPC environment. A typical build needs Fortran, C, MPI compiler wrappers, `make`, Python 3, and a Slurm runtime for the example submission scripts. The standalone component Makefiles use commands such as `gfortran`, `gcc`, `mpif77`, `mpif90`, and `mpicc`. For SPECSEM3D, select the serial Fortran, MPI Fortran, and C compilers with the `FC`, `MPIFC`, and `CC` arguments to `configure`. The integrated topography converter inherits this configured toolchain. Edit compiler variables directly only for components that do not use `configure`.

### 2.2 Build order

From the repository root that contains `src/DSM`, `src/InjectedWaves`, `src/Coupling`, and `src/SPECSEM3D`, build the components in this order:

```
cd src/DSM/src/DSM_Solver
make clean
make

cd ../DSM_FreqToTimeSac
make clean
make

cd ../../../../InjectedWaves/src
make clean
make

cd ../../Coupling/src
make clean
make

cd ../../SPECSEM3D
./configure FC=gfortran MPIFC=mpif90 CC=mpicc
make
```

The default SPECSEM3D target builds `xdecompose_mesh`, `xcubedsphere_topo`, `xmeshfem3D`, `xgenerate_databases`, and `xspecsem3D`. An MPI-enabled configuration also builds `xdecompose_mesh_mpi`. Step 0 later installs the selected case's model sources and rebuilds the canonical



**xgenerate\_databases**. The converter is therefore available to topographic cases without a separate demo-specific compilation step.

The converter is built locally with the compiler selected by **configure**; do not reuse a pre-compiled converter from a different Linux system. Its program entry point is **src/SPECFEM3D/src/cubedsphere\_topo/program\_cubedsphere\_topo.f90**; the implementation is split across **src/SPECFEM3D/src/cubedsphere\_topo/cubedsphere\_topo.f90** and companion Fortran sources. There is intentionally no single source file named **xcubedsphere\_topo.f90**. The Makefile in that source directory is a compatibility wrapper around the top-level SPECFEM3D build.

Ordinary SPECFEM3D **make clean** removes the converter with the other default targets. To force only the converter to be rebuilt after changing compilers or system libraries, run from **src/SPECFEM3D**:

```
make CLEAN=cubedsphere_topo clean
make xcubedsphere_topo
```

Then change into the selected example directory and run its Step 0 script:

```
./step0_prepare_*.sh --install
```

Each Step 0 selects a matching pair of Fortran source files. In the standardized DSM1D/DSM3D examples, it reads **DSM1D\_OR\_3D**. For DSM3D, **model\_tomography.f90** and **get\_model.F90** from the case-local **DATA/** directory are selected; for DSM1D, the shared pair in the bundled **1D\_DSM** subdirectory of **model\_tomo** is selected. Other model-specific cases install their case-local pair directly. The script copies both selected files into the SPECFEM3D **src/generate\_databases** directory, invokes **make** to rebuild the generator, and verifies the canonical **bin/xgenerate\_databases** executable. Model-suffixed database-generator executables are no longer selected by the examples.

The active Fortran pair and **xgenerate\_databases** are shared by all cases in one repository checkout. Rerun Step 0 for the intended case whenever you change cases, change **DSM1D\_OR\_3D**, or edit either Fortran source file. Do not prepare different cases concurrently against the same SPECFEM3D source tree.

## 2.3 Executables

| Executable                   | Purpose                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>dsmti</b>                 | DSM frequency-domain Green's function solver. Used for source-to-box, box-to-receiver, and 1-D reference synthetics.                           |
| <b>spectotime</b>            | Converts DSM frequency-domain coefficients to time-domain SAC files for 1-D synthetics.                                                        |
| <b>extract_Injectedwaves</b> | Interpolates DSM source-to-box Green's functions onto SEM boundary points, applies the selected time cut, and writes injected wavefield files. |
| <b>xcubedsphere_topo</b>     | Converts geographic interface topography to the cubed-sphere grid used by topographic SEM meshes.                                              |
| <b>xmeshfem3D</b>            | Builds the cubed-sphere local SEM mesh.                                                                                                        |
| <b>xgenerate_databases</b>   | Generates the SEM databases and applies the 1-D or 3-D model installed by Step 0 for the selected case.                                        |

|                                |                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>xspecfem3D</code>        | Runs the local SEM solver with the injected incident wavefield.                                                           |
| <code>coupling_integral</code> | Evaluates the representation integral using SEM boundary displacement/traction and DSM box-to-receiver Green's functions. |

---

## 2.4 Common compilation checks

After compiling the shared codes and running Step 0 for the selected case, return to the repository root and verify that the scripts can find the executables:

```
test -x src/DSM/src/DSM_Solver/dsmti
test -x src/DSM/src/DSM_FreqToTimeSac/spectotime
test -x src/InjectedWaves/src/extract_Injectedwaves
test -x src/Coupling/src/coupling_integral
test -x src/SPECFEM3D/bin/xcubedsphere_topo
test -x src/SPECFEM3D/bin/xmeshfem3D
test -x src/SPECFEM3D/bin/xgenerate_databases
test -x src/SPECFEM3D/bin/xspecfem3D
```

From the case directory, `./step0_prepare_*.sh --check` additionally verifies that both active Fortran files match the selected case and that `xgenerate_databases` is not older than those sources.

If a binary is installed in a nonstandard location, set the matching override in `DATA/Par_file_SEM_DSM`, for example `EXTRACT_INJECTEDWAVES_BIN` or `COUPLING_INTEGRAL_BIN`.

# Chapter 3

## DSM

### 3.1 Role of DSM

DSM supplies three different products:

1. Source-to-box Green's functions for the incident wavefield imposed on the SEM box.
2. Box-to-receiver Green's functions for the final representation integral.
3. Direct 1-D teleseismic synthetics, which are added to the scattered synthetics to form the final 3-D result.

All DSM runs must use a 1-D model consistent with the SEM background model. The workflow checks this consistency in the later SPECFEM3D and coupling steps by comparing the generated `dsm_model_input` with the DSM models stored under `WORK/DSM`.

### 3.2 Main DSM parameters

The DSM controls live in `DATA/Par_file_SEM_DSM`. The most commonly changed parameters are:

| Parameter                                                                                      | Meaning                                                                                                                                               |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>FREQ_RESOLVED</code>                                                                     | Highest frequency to resolve in the SEM box.<br>This also guides DSM sampling choices.                                                                |
| <code>DSM_NPROC</code>                                                                         | Requested MPI ranks for DSM solver jobs. The scripts cap this when the number of frequencies is smaller than the requested rank count.                |
| <code>DSM_TIME_LIMIT</code> , <code>DSM_MEM_PER_CPU</code><br><code>PHASE_SOURCE_TO_BOX</code> | Slurm resources for DSM jobs.<br>Target phase family used to estimate source-to-box arrival windows, for example <code>PKI</code> , <code>PK</code> . |
| <code>PHASE_BOX_TO_RECEIVER</code>                                                             | Target phase family used for the box-to-receiver Green's functions.                                                                                   |
| <code>TIME_LENGTH_DSM_SOURCE_TO_BOX</code>                                                     | Source-to-box DSM time length. It must cover the complete incident wavefield for the selected phase window.                                           |
| <code>TIME_LENGTH_DSM_BOX_TO_RECEIVER</code>                                                   | Box-to-receiver DSM time length.                                                                                                                      |

|                       |                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| TIME_LENGTH_DSM_1D    | Total time length for reference 1-D DSM synthetics. It should exceed the sum of the two leg lengths unless the scripts derive it automatically. |
| TELESEISMIC_COMPONENT | Receiver components to compute. Z runs vertical Green's functions; R or T also requires horizontal force components $F_r$ and $F_t$ .           |

---

### 3.3 Source modes

The parameter `SINGLE_FORCE_ENZ` controls the source representation:

| Value | Meaning                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------|
| -1    | Explosion source. The source-to-box run uses the <code>explosion</code> component.                         |
| 0     | Six moment-tensor Green's functions: $M_{zz}$ , $M_{rr}$ , $M_{tt}$ , $M_{zr}$ , $M_{zt}$ , and $M_{rt}$ . |
| 1     | Single force in the radial/north component $F_r$ .                                                         |
| 2     | Single force in the transverse/east component $F_t$ .                                                      |
| 3     | Single force in the vertical component $F_z$ .                                                             |

---

### 3.4 Running DSM source-to-box

Run:

```
./step4_dsm_src_to_box.sh
```

This script calls `auxiliary/prepare_src_to_box.py`, writes DSM input tables under `WORK/DSM/src_to_box/DATA`, creates one run directory per source component, copies `dsmti`, and submits `submit_DSM_*.sbatch`.

Important outputs are:

```
WORK/DSM/src_to_box/DATA/src_to_box_metadata.env
WORK/DSM/src_to_box/<source>/DATA/dsm_model
WORK/DSM/src_to_box/<source>/OUTPUT_FILES/
```

### 3.5 Running DSM box-to-receiver

Run:

```
./step5_dsm_box_to_recv.sh
```

This script calls `auxiliary/prepare_box_to_recv.py`. It prepares reciprocal DSM Green's functions from coupling-boundary points to the teleseismic receivers in `DATA/tele_station.txt`. The output components depend on `TELESEISMIC_COMPONENT`; for horizontal receiver components, both  $F_r$  and  $F_t$  are needed.

## 3.6 Running direct 1-D synthetics

Run:

```
./step6_1d_dsm_synthetics.sh
```

The script computes the reference 1-D waveform and converts frequency-domain DSM output to SAC. The final assembly step adds these traces to the 3-D scattered traces from the coupling integral.

## Chapter 4

# Injected Waves

### 4.1 Purpose

The injected-wave step converts DSM source-to-box Green's functions into the incident wavefield imposed on the outer SEM boundary. It performs four tasks:

1. Read the source-to-box DSM Green's functions.
2. Interpolate the DSM wavefield to SEM boundary points.
3. Transform frequency-domain DSM output to the time domain.
4. Cut a compact time window that fully contains the incident phase at all SEM-box corners.

### 4.2 Time-window control

The safest mode is to leave `START_TIME_CUT` and `END_TIME_CUT` blank. Then `step7_extract_injected_waves.sh` estimates the minimum and maximum target-phase arrival times at the eight SEM-box corners and applies `TIME_BUFFER_CUT_BEFORE` and `TIME_BUFFER_CUT_AFTER`. Use explicit cut times only when you have checked that the chosen window contains the entire incident wavefield.

The relevant parameters are:

| Parameter                                                                       | Meaning                                                                                                                           |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>START_TIME_CUT</code> , <code>END_TIME_CUT</code>                         | Optional explicit injected-wave window in seconds.                                                                                |
| <code>TIME_BUFFER_CUT_BEFORE</code>                                             | Buffer before the earliest box-corner target arrival.                                                                             |
| <code>TIME_BUFFER_CUT_AFTER</code>                                              | Buffer after the latest box-corner target arrival.                                                                                |
| <code>INJECTED_WAVES_NPROC</code>                                               | Requested MPI ranks. The script caps this by the SEM depth-table count because the extraction work is partitioned by depth index. |
| <code>NPT_ONE_PACKAGE</code>                                                    | Number of time samples handled per package.<br>Increase only if memory is sufficient.                                             |
| <code>INJECTED_WAVES_FILTERING</code>                                           | Whether to apply filtering during extraction.                                                                                     |
| <code>INJECTED_WAVES_FREQ_LOW</code> ,<br><code>INJECTED_WAVES_FREQ_HIGH</code> | Optional filter band.                                                                                                             |

### 4.3 Run command

After the SEM databases and source-to-box DSM jobs are finished, run:

```
./step7_extract_injected_waves.sh
```

The main outputs are:

```
WORK/InjectedWaves/DATA/injected_waves_metadata.env  
WORK/InjectedWaves/OUTPUT_FILES/
```

The metadata file records the cut window and derived timing. Later scripts use it to choose the SPECSEM3D solver duration when SPECSEM3D\_SOLVER\_NSTEP is not specified.

# Chapter 5

## SPECFEM3D

### 5.1 SEM box setup

The SEM box is a local cubed-sphere block centered on the target structure. The main geometric controls are:

| Parameter                          | Meaning                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------|
| CENTER_LATITUDE_IN_DEGREES         | Latitude of the SEM-box center.                                                               |
| CENTER_LONGITUDE_IN_DEGREES        | Longitude of the SEM-box center.                                                              |
| CENTER_DEPTH_KM                    | Center depth of the box.                                                                      |
| DEPTH_BLOCK_KM                     | Radial thickness of the box.                                                                  |
| GAMMA_ROTATION_AZIMUTH             | Rotation of the cubed-sphere grid.                                                            |
| ANGULAR_WIDTH_XI_IN_DEGREES        | Width in the local $\xi$ direction.                                                           |
| ANGULAR_WIDTH_ETA_IN_DEGREES       | Width in the local $\eta$ direction.                                                          |
| ES                                 |                                                                                               |
| NEX_XI, NEX_ETA, NZ_BOTTOM, NZ_TOP | Optional manual mesh-size overrides. Leave blank to let the scripts estimate from wavelength. |

The mesh resolution is estimated from the local wavelength  $\lambda = v/f_{\max}$ . In solid regions, the scripts use  $V_s$ ; in fluid regions, such as the outer core, they use  $V_p$ . Small-amplitude phases such as PKiKP are sensitive to SEM–DSM mismatch, so the example uses conservative settings such as `ELEMENTS_PER_WAVELENGTH = 1.6` and `OUTER_CORE_ELEMENTS_PER_WAVELENGTH = 3.0`.

### 5.2 Model modes

`DSM1D_OR_3D` chooses the SEM model:

- `DSM1D`: use the 1-D DSM background model. This is the preferred consistency test. The scattered wavefield should be close to zero except for numerical mismatch. Step 0 selects the bundled 1-D Fortran source pair.
- `DSM3D`: apply the case-specific 3-D model implemented by `DATA/model_tomography.f90` and `DATA/get_model.F90`.

Step 0 copies the selected pair into the `SPECFEM3D src/generate_databases` directory as `model_tomography.f90` and `get_model.F90`, then rebuilds `bin/xgenerate_databases`. For



a new or changed 3-D model, edit the case-local pair and rerun Step 0 with `--install` before regenerating the SEM databases.

### 5.3 Prepare and generate the mesh

Run:

```
./step0_prepare_*.sh --install
./step1_specfem_mesh_database.sh
./step2_3Dmodel_implementation.sh
./step3_submit_specfem_mesh_gen.sh
```

**step0** installs both selected Fortran files, rebuilds `xgenerate_databases`, and prepares any case-specific structural inputs. **step1** prepares `WORK/SPECFEM3D/DATA`, calculates mesh settings, copies templates, writes `Par_file`, and generates the DSM model used by the SEM case. **step2** reports whether the case is using DSM1D or DSM3D. **step3** verifies the active source pair and executable, then writes and submits the Slurm jobs for `xmeshfem3D` and the canonical `xgenerate_databases`.

For interface-topography cases, the applicable DSM3D path in Step 1 invokes `xcubedsphere_topo`; flat or DSM1D modes skip that conversion. In cases that generate injected waves locally, Step 1 only records the future `INJECTED_WAVEFIELD_PATH` in the SPECFEM3D parameter file; Step 7 later creates `WORK/InjectedWaves`. Cases configured specifically to reuse an existing injected-wave product require that product to exist and may validate it during Step 1.

Stochastic-heterogeneity generators need a provisional effective SEM GLL-point spacing before the databases exist. Most cases read it from `HETEROGENEITY_DX_EFF_M`; some generators define `dx_eff` locally. After the `xgenerate_databases` job submitted by Step 3 finishes successfully, compare the configured value with `*** Max GLL point distance` in `WORK/SPECFEM3D/OUTPUT_FILES/output_mesher.txt`. If they differ significantly, update the configured value, rerun `generate_stochastic_heterogeneity_tiled_overlap.py` with `DATA/heterogeneities` as its working directory, then change to `WORK/SPECFEM3D` and resubmit `run_generate_databases.sbatch`. The existing mesh can be reused; Step 1 and `xmeshfem3D` need not be rerun unless a mesh-setting parameter was also changed.

Before continuing, check that the database directory exists:

```
ls WORK/SPECFEM3D/OUTPUT_FILES/DATABASES_MPI
```

### 5.4 Run the coupled SEM solver

After the injected-wave files are available, run:

```
./step8_specfem_solver.sh
```

This script verifies DSM-model consistency, updates the solver `DT` and `NSTEP`, points SPECFEM3D to the injected wavefield, and submits `xspecfem3D`. If `SPECFEM3D_SOLVER_DT` is blank, the script uses the mesher's suggested time step, multiplies it by `SPECFEM3D_SOLVER_DT_SAFETY_FACTOR`, and optionally caps it with `SPECFEM3D_SOLVER_MAX_DT`. For difficult injected-wave runs, set `SPECFEM3D_SOLVER_DT` explicitly.

The solver writes displacement and traction on the coupling boundary. These files are consumed by `step9_coupling.sh`.

## Chapter 6

# Coupling

### 6.1 Representation integral

The coupling step evaluates Eq. 1.1. It combines:

- SEM displacement and traction saved on the internal coupling boundary.
- DSM box-to-receiver Green's functions and spatial derivatives.
- Boundary quadrature weights, normals, Jacobians, and coordinate transforms.

The coupling boundary should be placed inside the SEM box, away from the outer absorbing boundary and away from the strongest local heterogeneity when possible. In practice, most numerical error comes from the top and bottom coupling faces because the injected teleseismic body-wave field usually has a strong vertical component. Side-face contributions are often smaller and may partly cancel because opposite faces have opposite outward normals.

### 6.2 Resolution of the surface integral

`LOW_RESOLUTION = .false.` saves all GLL points on coupling faces and is the recommended production setting. `LOW_RESOLUTION = .true.` saves only the central GLL point on each coupling face. This is faster and smaller on disk but can produce much larger errors for high-frequency phases such as PKiKP.

The coupling integral uses interpolation orders `COUPLING_NFIT_DEP` and `COUPLING_NFIT_DIST`; values of 4 are a typical starting point. Keep these even and avoid changing them unless you are testing interpolation error.

### 6.3 Run command

Run:

```
./step9_coupling.sh
```

This script prepares `WORK/Coupling/DATA`, verifies that the DSM models used by the reusable products match the current case, creates the coupling parameter file, and submits `coupling_integral`. Important controls are:

| Parameter                    | Meaning                                                                     |
|------------------------------|-----------------------------------------------------------------------------|
| COUPLING_SEM_OUTPUT_DIR      | Optional path to a reusable WORK/SPECFEM3D/OUTPUT_FILES directory.          |
| COUPLING_DSM_BOX_TO_RECV_DIR | Optional path to reusable box-to-receiver DSM products.                     |
| COUPLING_NPOINTS_TAPER_SEM   | Number of SEM time samples tapered at the start and end before convolution. |
| COUPLING_NPROC               | Requested MPI ranks for the integral.                                       |
| COUPLING_MEM_PER_CPU         | Slurm memory per CPU. The coupling integral can be memory intensive.        |

## 6.4 Assemble final synthetics

The coupling result is the scattered waveform. Add it to the direct 1-D DSM waveform with:

```
./step10_assemble_3d_sem_dsm_synthetics.sh
```

The helper script `auxiliary/assemble_3d_sem_dsm_synthetics.py` chooses the correct 1-D DSM SAC directory from `SINGLE_FORCE_ENZ` and writes the final 3-D SEM–DSM synthetics.

## 6.5 Quality-control checklist

For each new case, run a DSM1D test before interpreting the 3-D model. The scattered waveform should be small compared with the target phase. If it is not, check:

- The DSM model used by source-to-box, box-to-receiver, and SEM database generation is identical.
- The injected-wave cut window fully covers the incident phase.
- DT is small enough for the injected-wave run.
- The coupling boundary is not too close to the absorbing boundary or a sharp discontinuity.
- `LOW_RESOLUTION` is set to `.false.` for production.
- The SEM mesh has enough elements per wavelength in the slowest region.

# Chapter 7

## Examples

### 7.1 General example workflow

The examples all follow the same pattern:

```
# edit DATA/Par_file_SEM_DSM, DATA/CMTSOLUTION, DATA/tele_station.txt
./step0_prepare_*.sh --install
./step1_specfem_mesh_database.sh
./step2_3Dmodel_implementation.sh
./step3_submit_specfem_mesh_gen.sh

./step4_dsm_src_to_box.sh
./step5_dsm_box_to_recv.sh
./step6_1d_dsm_synthetics.sh

# after the mesh/database and source-to-box DSM jobs finish
./step7_extract_injected_waves.sh
./step8_specfem_solver.sh

# after the SEM solver and box-to-receiver DSM jobs finish
./step9_coupling.sh
./step10_assemble_3d_sem_dsm_synthetics.sh
```

Step 0 must finish before the SEM mesh/database workflow. Run it again after switching cases or DSM1D\_OR\_3D, or after changing either case-local Fortran source file. Step 3 rejects an active source pair or `xgenerate_databases` executable that does not match the current case. Steps 4, 5, and 6 can usually run independently after the parameter files are ready. Step 7 requires both source-to-box DSM output and SEM databases. Step 8 requires the injected wavefield. Step 9 requires the SEM solver output and box-to-receiver DSM output.

**Frequency range and post-processing.** To keep the computational cost of the examples manageable, the current demo cases are configured for relatively low frequencies. The final 3-D synthetics may need to be band-pass filtered in a frequency range appropriate for the case before the 3-D effects are clearly visible. For example, the raw 3-D synthetics from the PcP heterogeneity case can contain substantial high-frequency noise; a 0.2–0.4 Hz band-pass is a useful starting point for displaying its 3-D signal. Adjust the band for the model and phase being studied, and apply the same filter to the paired 1-D and 3-D synthetics when comparing them.

## 7.2 PKIKP with inner-core rotation

This example models PKIKP propagation through a heterogeneous inner core and compares equatorial and polar paths with different rotations of the 3-D heterogeneity. The example directories are organized around paths such as:

```
example/PKIKP_IC_Rotation_Hete_demo/EquatorialPath
example/PKIKP_IC_Rotation_Hete_demo/EquatorialPath_1.5degRotated_DSM_database_existing
example/PKIKP_IC_Rotation_Hete_demo/PolarPath_DSM_database_existing
example/PKIKP_IC_Rotation_Hete_demo/PolarPath_1.5degRotated_DSM_database_existing
```

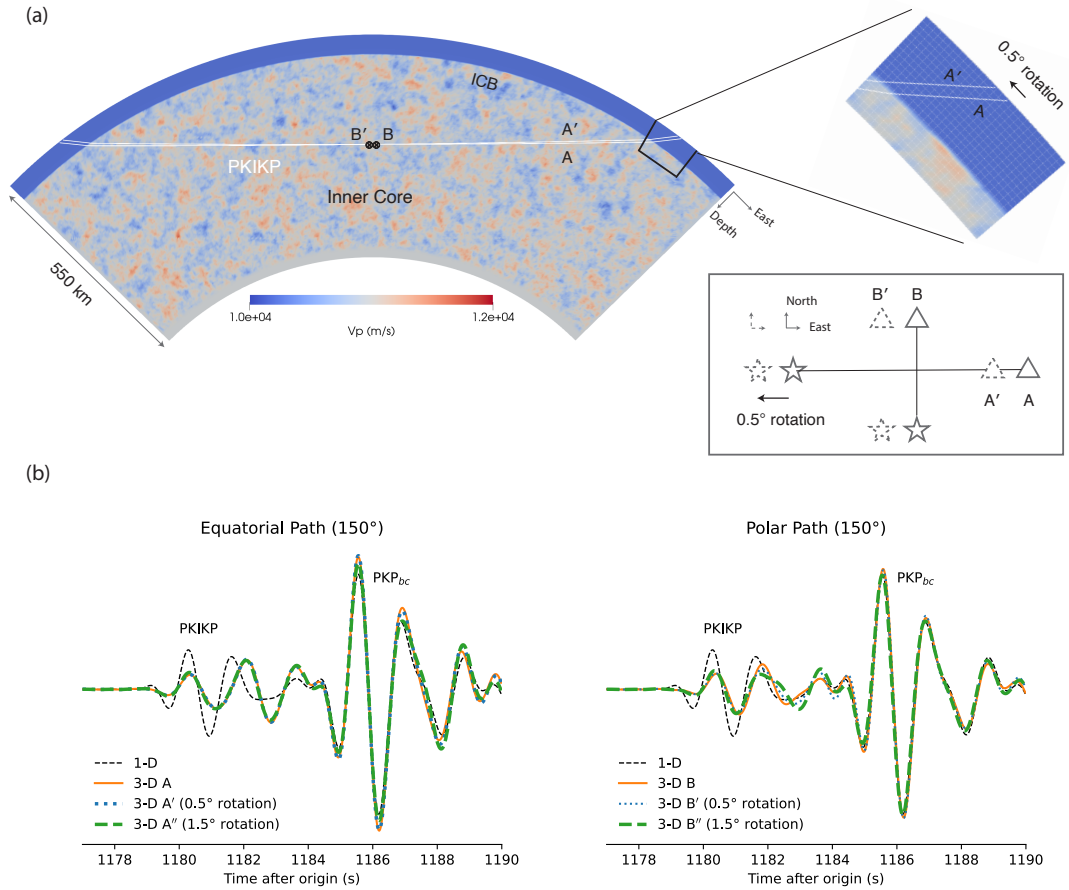


Figure 7.1: PKIKP application with rotated inner-core heterogeneity. The rotated model changes the 3-D structure sampled by the PKIKP path while preserving the same reference 1-D propagation outside the SEM box.

### 7.2.1 Recommended setup

Use a box centered in the inner core along the PKIKP path. Set `PHASE_SOURCE_TO_BOX` and `PHASE_BOX_TO_RECEIVER` to include the internal-core phase family needed by the corner-arrival estimator, for example PKI, PK. For initial checks, set `DSM1D_OR_3D` = `DSM1D`. For the rotation experiment, switch to `DSM3D`, edit the case-local `model_tomography.f90` and `get_model.F90` pair

if needed, run `./step0_prepare_*.sh --install`, and rerun the database, solver, coupling, and assembly steps.

### 7.2.2 Reusing DSM products

For rotation tests, the 1-D source-to-box and box-to-receiver DSM products can often be reused when the source, receivers, 1-D model, SEM-box geometry, and time windows are unchanged. The reusable-product controls are:

```
DSM_SRC_TO_BOX_DIR =
DSM_BOX_TO_RECV_DIR =
INJECTED_WAVEFIELD_PATH =
INJECTED_WAVES_METADATA =
COUPLING_DSM_BOX_TO_RECV_DIR =
```

Keep these blank for a clean end-to-end run. Set them only when you have verified that the current case geometry and DSM model are identical to the saved product.

### 7.2.3 Interpreting output

The useful diagnostic comparison is between the DSM1D result and the DSM3D rotated-model result. The difference isolates the waveform effect of inner-core heterogeneity and its rotation. Also compare equatorial and polar paths, because the same rotated heterogeneity can affect them differently.

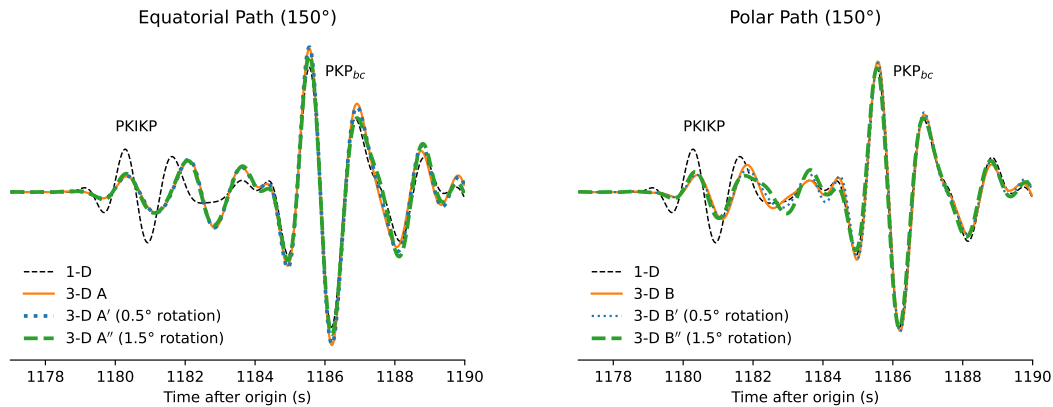


Figure 7.2: Example comparison of equatorial and polar PKIKP paths. Use this type of comparison to separate path geometry from the imposed rotation of inner-core structure.

## 7.3 PKiKP with ICB topography

This example models PKiKP reflections from inner-core boundary topography. The example directories include:

```
example/PKiKP_ICB_topo_demo/Explosion_demo
example/PKiKP_ICB_topo_demo/Explosion_DSM_database_existing_demo
```

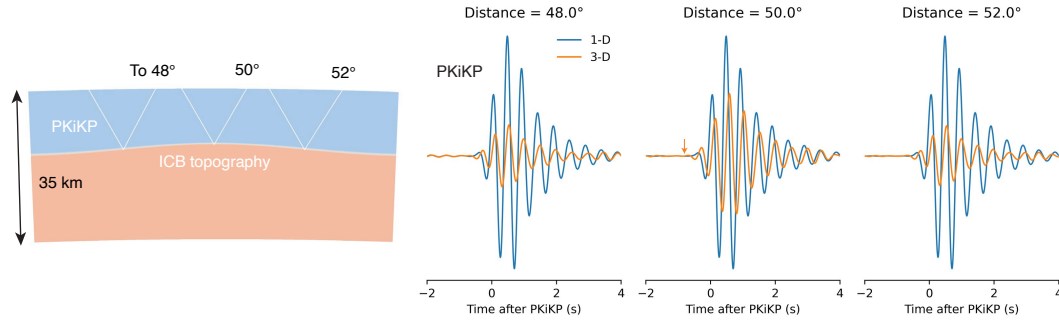


Figure 7.3: PKiKP application with ICB topography. The SEM box surrounds the reflection region, while the source-side and receiver-side propagation are handled by DSM in the 1-D background model.

### 7.3.1 Topography preparation

The PKiKP example uses an ICB topography file such as:

```
DATA/topo_files/latlon_ICB_topo.txt
```

The parameter `ICB_TOPOGRAPHY_FILE` points to this file. During database preparation, the topography is converted to the cubed-sphere interface format with `xcubedsphere_topo`. Build it from the repository root with `make -C src/SPECFEM3D xcubedsphere_topo`.

After setting `DSM1D_OR_3D`, run `./step0_prepare_model.sh --install`. In addition to installing the two Fortran sources and rebuilding `xgenerate_databases`, the `DSM3D` path generates the ICB topography input before mesh preparation.

### 7.3.2 Parameter choices

PKiKP is small, so numerical mismatch can be visible. Use the high-resolution coupling boundary:

```
LOW_RESOLUTION = .false.
```

Use conservative mesh settings near the ICB and test `DSM1D` before interpreting topography. The example uses a relatively high `OUTER_CORE_ELEMENTS_PER_WAVELENGTH` because the coupling includes waves that propagate through the fluid outer core and reflect near the ICB.

### 7.3.3 Suggested run sequence

For a clean PKiKP topography run:

```
# 1-D consistency check
set DSM1D_OR_3D = DSM1D in DATA/Par_file_SEM_DSM
./step0_prepare_model.sh --install
./step1_specfem_mesh_database.sh
./step3_submit_specfem_mesh_gen.sh
./step4_dsm_src_to_box.sh
./step5_dsm_box_to_recv.sh
./step6_1d_dsm_synthetics.sh
./step7_extract_injected_waves.sh
./step8_specfem_solver.sh
```

```

./step9_coupling.sh
./step10_assemble_3d_sem_dsm_synthetic.sh

# 3-D ICB topography run
set DSM1D_OR_3D = DSM3D in DATA/Par_file_SEM_DSM
./step0_prepare_model.sh --install
./step1_specfem_mesh_database.sh
./step3_submit_specfem_mesh_gen.sh
./step8_specfem_solver.sh
./step9_coupling.sh
./step10_assemble_3d_sem_dsm_synthetic.sh

```

When the source, stations, 1-D model, and SEM-box geometry are unchanged, the DSM products from the 1-D check can be reused for the 3-D topography run.

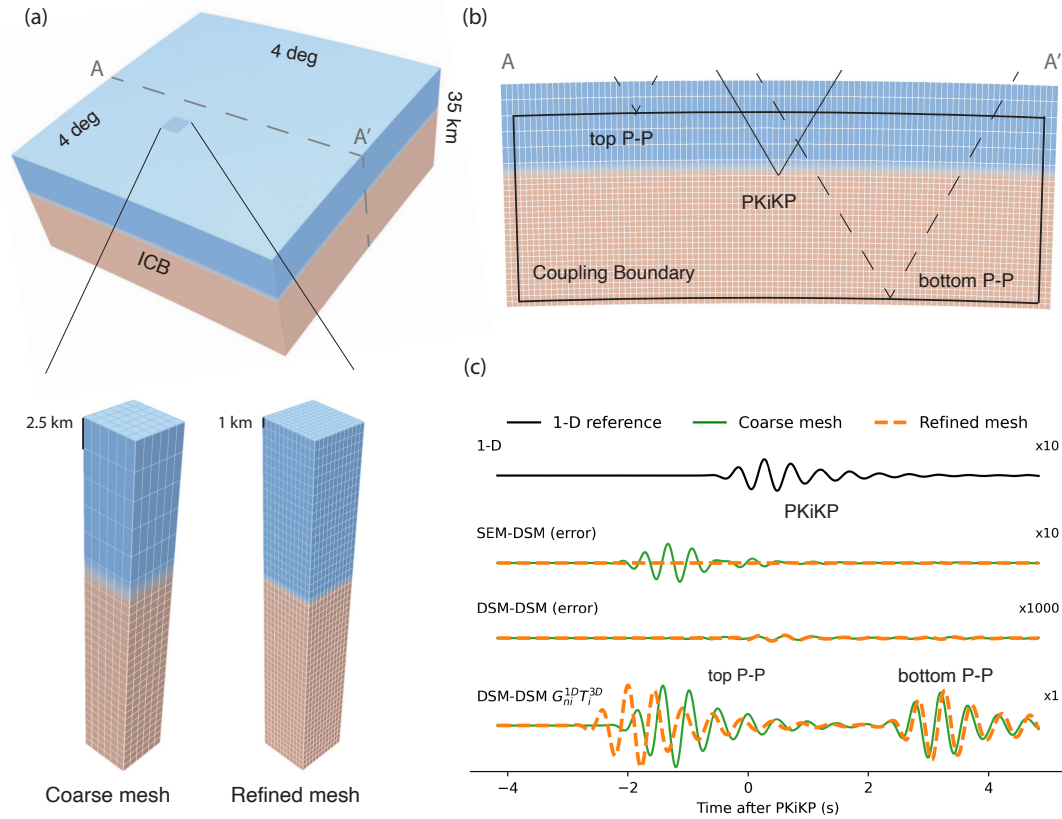


Figure 7.4: PKiKP 1-D benchmark. This type of test is used to quantify SEM–DSM numerical mismatch before adding ICB topography.

## 7.4 Output organization

Although examples differ in model details, the final products usually appear under:

```

WORK/Coupling/OUTPUT_FILES/
WORK/DSM/1D_tele_syn/
OUTPUT_FILES_DSM1D_*

```



```
OUTPUT_FILES_DSM3D_*
```

Use the DSM1D and DSM3D directories as paired products: the DSM1D run quantifies numerical error and the DSM3D run contains the target structural signal plus the same numerical background.

# Troubleshooting

| Symptom                                                                       | Likely fix                                                                                                                                                                                 |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>dsmti</code> not found                                                  | Build <code>src/DSM/src/DSM_Solver</code> or correct the relative path in the case scripts.                                                                                                |
| <code>extract_Injectedwaves</code> not found                                  | Build <code>src/InjectedWaves/src</code> or set <code>EXTRACT_INJECTEDWAVES_BIN</code> .                                                                                                   |
| <code>coupling_integral</code> not found                                      | Build <code>src/Coupling/src</code> or set <code>COUPLING_INTEGRAL_BIN</code> .                                                                                                            |
| <code>xcubedsphere_topo</code> is missing or reports a GLIBC version error    | Configure SPECSEM3D for the local toolchain, then run <code>make -C src/SPECSEM3D CLEAN=cubedsphere_topo o clean</code> followed by <code>make -C src/SPECSEM3D xcubedsphere_topo</code> . |
| Injected-wave step stops because SEM depth tables are missing                 | Wait for <code>xmeshfem3d</code> and <code>xgenerate_databases</code> to finish successfully.                                                                                              |
| <code>xgenerate_databases</code> is missing, stale, or built for another case | Run <code>./step0_prepare_*.sh --install</code> from the intended case directory, then rerun Step 3.                                                                                       |
| SPECSEM3D solver overflows                                                    | Reduce <code>SPECSEM3D_SOLVER_DT</code> , reduce the injected-wave frequency band, or inspect the injected-wave cut window.                                                                |
| Large scattered waveform in DSM1D mode                                        | Check DSM-model consistency, mesh resolution, coupling-boundary placement, and <code>LOW_RESOLUTION</code> .                                                                               |
| Coupling job runs out of memory                                               | Increase <code>COUPLING_MEM_PER_CPU</code> , reduce the receiver/component set, or split the coupling job.                                                                                 |
| Final 3-D synthetics are missing the reference phase                          | Run <code>step6_1d_dsm_synthetics.sh</code> and then rerun <code>step10_assemble_3d_sem_dsm_synthetics.sh</code> .                                                                         |